NFL DFS Data Inventory

Everything the project has, what it could get for free, and what does not exist anywhere.
Project: `C:\new_career2026\nfl-dfs` · Built from the files on disk, 2026-09-15.

How to read this

Every dataset carries one of four labels:

Label	Meaning
IN MODEL	On disk and already feeding the points model (`src/features.py`).
ON DISK	Downloaded, but the model has never used it. **This is where most of the untapped value is.**
NEW, UNUSED	Free data downloaded 2026-09-15 with `python src/fetch_nflverse.py --extras-only --end 2026`. Not in the model yet.
MISSING	Could not be found free or in the FantasyData subscription.

The whole picture

Dataset	Grain	Seasons	Rows	Status
Weekly player stats (nflverse)	player-week	2016–2024	49,161	IN MODEL
Schedules + Vegas lines + weather	game	2016–2025	2,761	IN MODEL
Snap counts	player-week	2016–2025	253,106	IN MODEL
Model table (the features)	player-week	2016–2024	45,715	IN MODEL
Injury reports	player-week	2016–2025	55,556	ON DISK
Depth charts	team-week / daily	2016–2025	886,389	ON DISK
Player ID crosswalk + age, draft, size	player	current	12,492	ON DISK
DK salaries (RotoGuru)	player-week	2016–2021	41,201	ON DISK
DK salaries + projection (FantasyData)	player-week	2022–2025	72 files	ON DISK
DFS projections export (FantasyData)	player-week	scattered 2017–2026	73 files	ON DISK
Advanced metrics (FantasyData)	season total	2021–2025	20 files	ON DISK
Efficiency metrics (FantasyData)	season total	2021–2025	20 files	ON DISK
Live DK slate snapshots	player-snapshot	2026 Wk 1	4 files	ON DISK
Play-by-play	play	1999–2026	~48k plays/yr	NEW, UNUSED
Participation (who was on the field)	play	2016–2025	~48k plays/yr	NEW, UNUSED
FTN film charting	play	2022–2026	~47k plays/yr	NEW, UNUSED
Next Gen Stats (separation, cushion)	player-week	2016–2026	14,877 (rec.)	NEW, UNUSED
Pro Football Reference advanced	player-week	2018–2026	—	NEW, UNUSED
Draft picks · Combine · Contracts	player	historical	—	NEW, UNUSED
Historical ownership, line movement, player props…	—	—	—	MISSING

Three words used throughout

Term	Plain English

Grain	What one row means. "Player-week" = one row per player per game. "Season total" = one row per player per season. The model needs player-week.
Known before kickoff	The one rule for any feature: could you have known it before the game started? If not, the model is peeking at the answer (*leakage*). A season total used for a game *inside* that season breaks this rule; last season's total does not.
Fill %	The share of rows where a column actually has a value. 65% means 35% are blank.

1 · In the model today

Four sources feed the model. It ranks each week's players with a score of **0.5958** (slate-rank Spearman, 2021–2024) against **0.5791** for a plain rolling average. An honest caveat: those two numbers were measured on different row counts (21,071 vs 16,751), so the true gap is not settled yet.

Weekly player stats IN MODEL

`data/raw/weekly.parquet` · 49,161 rows · 53 columns · 2016–2024 · QB/RB/WR/TE · source: nflverse

The box score for every player in every game. The model's target (DraftKings points) is computed from it.

Column group	What it holds	Fill
Passing	completions, attempts, yards, TDs, INTs, sacks, air yards, yards after catch, first downs	100%
Rushing	carries, yards, TDs, fumbles, first downs	100%
Receiving	receptions, targets, yards, TDs, air yards, yards after catch, first downs	100%
Shares	`target_share` (his share of team targets), `air_yards_share`, `wopr` (a blend of the two)	81%
Efficiency (EPA)	`passing_epa`, `rushing_epa`, `receiving_epa`: expected points added (how much a player's plays moved the team's scoring odds)	12–81%

Gap: stops at 2024. The 2025 season has to be rebuilt from play-by-play (free) before 2025 can be trained on.

Schedules, Vegas lines, weather IN MODEL

`data/raw/schedules.parquet` · 2,761 games · 46 columns · 2016–2025 · source: nflverse

Columns	Plain English	Fill
`spread_line`, `total_line`, moneylines, odds	One Vegas line per game (no timestamp, so no line movement)	100%
`temp`, `wind`, `roof`, `surface`	Weather. Blanks are domes, which the code fills as 70° with no wind.	65%
`away_rest`, `home_rest`, `div_game`	Days since last game; divisional matchup	100%
`away_qb_id`, `home_qb_id`	Starting QB each game (drives the `green_qb` feature)	100%
`home_coach`, `away_coach` UNUSED	Head coach (95 distinct). Could capture coaching changes and play-calling tendencies.	100%
`referee` UNUSED	Referee crew chief (33 distinct). Crews differ in penalty rates, which changes the number of plays.	100%
`home_score`, `away_score`	Final score. **Never a feature**: not known before kickoff.	100%

Snap counts IN MODEL

`data/raw/snaps.parquet` · 253,106 rows · 2016–2025 · all positions incl. defense and special teams · source: Pro Football Reference via nflverse

`offense_pct` = the share of his team's offensive plays he was on the field for. The model uses 3- and 5-game averages. Joined by name + team (it uses a different player ID), so a few players get lost in the join.

The model table IN MODEL

`data/processed/model_table.parquet` · 45,715 player-weeks · 2016–2024 · built by `src/features.py`

Family	Features (39 total)
Recent scoring	`dkp_r3 / r5 / r8` (averages), `dkp_ewm` (recent-weighted), `dkp_lag1` (last game), `dkp_std_r8` (how streaky)
Opportunity	target share, air-yards share, WOPR, targets, carries, receptions, snap %, targets per snap, games so far
Vegas / context	implied team & opponent totals, total, spread, home, rest, divisional, green QB
Matchup	`def_allowed_r4 / r8`: points this defense gave up to his position recently
Weather	wind, temp, indoor
Interactions	`spread_x_rb`, `spread_x_pass` (game-script effects)
Position	QB / RB / WR / TE flags

Built-in blind spot: every rolling average resets each season, so in Week 1 the model knows nothing about any player's history. **Nothing about injuries, depth, age, price, or routes is in here yet.**

2 · On disk, never used

Already downloaded. Turning any of these into features costs code, not money.

Injury reports ON DISK

`data/raw/injuries.parquet` · 55,556 rows · 2016–2025 · ~5,100–6,200 per season · source: official NFL reports via nflverse

Column	Values seen
`report_status` (game designation)	Questionable 14,875 · Out 10,257 · Doubtful 1,672 (48% filled; blank = practiced but not designated)
`practice_status`	Full 25,719 · Did Not Participate 15,654 · Limited 13,925
`report_primary_injury`	Body part (hamstring, knee, …)
`date_modified`	When the report was last updated (89% filled): a rough clock on when news became public

Feature #1 idea: when a starter is Out, how many targets and carries is he leaving behind, and who inherits them?

Depth charts ON DISK

`data/raw/depth_charts.parquet` · 886,389 rows · source: team-published, via nflverse

Seasons	Format
2016–2024	One chart per team per week (332,174 rows, regular season + playoffs). `depth_team` 1/2/3 = starter / backup / third.
2025	**Daily** copies, 219 days from Aug 3 2025 to Mar 14 2026 (554,215 rows). Different columns (`pos_rank`, `pos_slot`) and **no season/week column**, so dates must be matched to weeks.

Teams publish these themselves and they are often stale or vague. Pair them with injury reports: injuries say who is *out*, depth charts say who is *next*.

Player ID crosswalk ON DISK

`data/raw/ids.parquet` · 12,492 players · 35 columns · source: DynastyProcess via nflverse

Translates one site's player ID into another's. **It is also the only place the project has age and draft data.**

Column	Use	Fill
`gsis_id`	The NFL's ID; what weekly stats use	64%
`fantasy_data_id`	Links FantasyData files. **Matches 294 of the top 300 salaried players** (1,101 of 1,391 skill players overall)	47%
`birthdate`, `age`	Aging curves. **No age feature exists in the model.**	90%
`draft_round`, `draft_pick`	Draft capital: how much a team invested in him, which is a signal for rookies with no stats	58%
`height`, `weight`, `college`	Body type (e.g., size for goal-line backs)	91–95%

DraftKings salaries, 2016–2021 ON DISK

`data/raw/dk_salaries.parquet` · 41,201 rows · weeks 1–17 (2021 Week 18 missing) · source: RotoGuru

Salary **and actual DK points** in the same row, so points per dollar is directly measurable. QB 3,804 · RB 10,800 · WR 14,849 · TE 8,676 · DEF 3,072 rows. Names are "Last, First" format; team codes are lowercase 3-letter (`kan`, `sfo`).

2 · On disk, never used (continued): FantasyData

Bought one month of All Access and canceled on 2026-09-15. Folder: `data/raw/fantasydata/`.

Weekly salaries + FantasyData projection ON DISK

`salaries/dk_{season}_wk{NN}.csv` · 72 files · 2022–2025, weeks 1–18, complete · ~650–950 players/week

Column	Meaning
`salary_draftkings`	DK salary
`fpts_draftkings`	**FantasyData's projection**, not actual points (equals actual only 2.8% of the time; correlation with actual 0.65). **This is the benchmark to beat.**
`id`	FantasyData player ID → crosswalk → NFL ID
`opp_rank`, `opp_pos_rank`	FantasyData's defense rankings

- Includes kickers, defenses, individual defensive players, and many backups who never play.
- Salaries agree with RotoGuru on **99.7%** of 384 overlapping players (2021 Week 1), so the two sources can be combined.
- 2022 Week 16 (Christmas): 7% of salaries differ from the projections export, because different DK slates were used.
- Files had no season/week column; each was identified by matching its games to the NFL schedule (all 72 matched exactly).

DFS projections export ON DISK

`projections/dk_{season}_wk{NN}.csv` · 73 files

Season	Weeks on disk	Season	Weeks on disk
2017	1–17 except 16	2021	1–18 (all)
2018	1–17 (all)	2022	2–18
2019, 2020	Week 1 only	2023	17–18
		2026	Week 1

Same salary and projection as the salary files, plus `roster_slots` and `own_pct`. **Ownership exists only for 2021 Weeks 1–5** (167–267 players per week, adding up to ~100% per roster slot); blank everywhere else. Holiday weeks sometimes leave out Thanksgiving or Saturday games.

Advanced metrics, season totals ON DISK

`advanced/{qb,rb,wr,te}_{2021..2025}.csv` · 20 files

Pos	Columns
WR/TE	**RouteParticipation** (share of pass plays he ran a route), SnapShare, AverageTargetDistance, AirYards, RedZoneTargets, EndzoneTargets, TargetShare, FantasyPointsPerTarget, ~~TargetAccuracy~~ (dead in 2024–25)
RB	SnapShare, OpportunityShare, ShotgunCarryRate, UnderCenterCarryRate, AverageDefendersInTheBox, Stacked/Base/LightFrontCarryRate, ~~Shotgun/UnderCenter YPC~~ (junk 2021–22). **No route participation.**
QB	**MoneyThrows**, ProtectionRate, PressuredCompletionPercentage, AirYardsPerAttempt, DeepBallAttempts, DeepBallCompletionPercentage

Fill improves every year: the 2021 files are 50–77% filled (blanks are fringe players), 2024–25 are 93–100%.

Efficiency metrics, season totals ON DISK

`efficiency/{qb,rb,wr,te}_{2021..2025}.csv` · 20 files · nearly 100% filled

Pos	Columns (**bold** = judged from game film, only source)
RB	**JukeRate, EvadedTackles, YardsCreated** (+ per game, per carry), BreakawayRuns/Rate, Drops, DropRate, points per carry/target/opportunity
WR/TE	**ContestedTargets, ContestedCatches, ReceiverContestedCatchRate, TrueCatchRate**, AverageCushion, TargetSeparation, YardsPerRouteRun, Drops
QB	**InterceptablePasses, TruePasserRating**, PlayactionPassCompletionPercentage, ReceiverTargetSeparation, RedZoneAttempts, DroppedPasses, ~~AccuracyRating~~ (dead 2024–25)

Season totals can only describe the **next** season (known-before-kickoff rule). Five seasons = four year-over-year comparisons: enough for a research test, thin for training a model.

Live DraftKings slate snapshots, 2026 Week 1 ON DISK

`data/slates/dg151307/` (12-game main slate, 744–746 players) · `data/slates/dg153069/` (711 players) · pulled Sept 8 and Sept 10

Columns: salary, `status` (OUT/Q/IR), `news_status` (breaking news flag), `dk_fppg`, `dk_oprk`, kickoff time, dome, plus a projections file with the model's Week 1 numbers. **No snapshot from Sunday morning**, so last-minute status changes weren't captured. Only going forward can this build a salary and news history.

3 · Free data, downloaded 2026-09-15

All free from nflverse, now in `data/raw/`. None of it is in the model yet. Current-season (2026) files grow every week; re-run the fetch script with `--refresh` to update them.

Play-by-play NEW, UNUSED

`data/raw/pbp/` · 2016–2026 (2026 = Week 1 so far) · ~47–50k plays a season · 372 columns · 199 MB

Every play: down, distance, field position (`yardline_100`), pass/run, air yards, who was targeted, result, win probability, EPA. Anything situational can be counted from it:

- **Red-zone and goal-line** carries and targets (inside the 20, 10, 5)
- **End-zone targets** and deep targets
- **Pass rate over expected** by team (does a coach throw more than the situation calls for?)
- **Pace**: seconds per play, plays per game
- **2025 weekly stats**, rebuilt to fill the gap

Participation NEW, UNUSED

`data/raw/participation/` · 2016–2025 (2026 not published yet) · ~45–51k plays a season · player names and positions added from 2023

The IDs of all 11 offensive and 11 defensive players on the field, plus formation, personnel, defenders in the box, pass rushers, time to throw, pressure, man/zone coverage, and a route label on pass plays.

Weekly routes run, calculated free: the share of team pass plays a receiver was on the field for. Checked against FantasyData's 2025 route participation, it landed within 1–4 points for six top WRs (e.g., Chase 96.5% vs 97.2%, Nacua 78.7% vs 77.2%). Not yet tested for TEs and RBs, where blocking snaps make it looser.

FTN film charting NEW, UNUSED

`data/raw/ftn/` · 2022–2026 · ~42–48k plays a season

Human-charted flags on every play: `is_catchable_ball`, `is_contested_ball`, `is_created_reception`, `is_drop`, `is_interception_worthy`, `is_play_action`, `is_motion`, `is_rpo`, `is_screen_pass`, `n_blitzers`, `is_qb_out_of_pocket`, `is_throw_away`.

This is the weekly, free version of several FantasyData film stats (contested catches, catchable-ball rate, interception-worthy throws).

Next Gen Stats (weekly) NEW, UNUSED

`data/raw/ngs/` · 2016–2026 · receiving 14,877 · rushing 6,127 · passing 5,997 rows (week 0 = season totals)

Receiving: `avg_separation` (distance from the nearest defender at the catch or incompletion), `avg_cushion` (how far off the defender plays at the snap), `avg_intended_air_yards`, `avg_yac_above_expectation`. Only the more-involved players each week (~1,350 receiving rows a season). **Cushion is the defense's own read on a receiver's speed**, making it the best free "losing a step" signal.

Pro Football Reference advanced (weekly) NEW, UNUSED

`data/raw/pfr/` · rush, rec, pass, def · 2018–2026 (2026 Week 1 incomplete upstream: 4 teams)

Rushing yards before and after contact, broken tackles, receiving drops and drop %, passing drops, and more. Broken tackles run about **one-third** of FantasyData's "evaded tackles" (a stricter definition), but in six 2025 backs the top two and bottom one were the same players on both lists.

Draft picks · Combine · Contracts NEW, UNUSED

`data/raw/players/` · draft picks 12,927 (1980–2026) · combine 8,968 (2000–2026) · contracts 52,816

Full draft history; 40-yard dash, weight, vertical, etc.; contract value and guaranteed money. A team that just gave a player guaranteed money has a reason to keep using him.

4 · What is NOT there

Searched for and not found free or in the subscription, or only partly available. Each gap is also a clue: if data is hard to get, the market may price it poorly.

Missing	Why it matters	Closest thing you have
Historical ownership MISSING	Tournaments are won with players the field ignores. You can't learn what the crowd overlooks without knowing what it picked.	2021 Weeks 1–5 only. Could be approximated: ownership tends to follow salary value and recent points.
Line movement MISSING	How the betting market reacted to news during the week	One line per game, two sources that differ slightly, no timestamps
Player props MISSING	Vegas's own projection for each player (e.g., receiving yards over/under). Likely the sharpest projection that exists.	Team totals only (implied team total)
Contest results MISSING	What winning lineups scored, field sizes, payout structures. Decides how much risk a lineup should take.	Your own DK entry history (downloadable from your account)
Precipitation MISSING	Rain and snow hypotheses	Temp and wind only; would need Meteostat/NOAA joined on stadium location
News timing MISSING	An injury announced Friday vs. 90 minutes before kickoff gets priced very differently	Injury `date_modified` (rough); DK `news_status` going forward
Sunday inactives (history) MISSING	The final "who is really playing" list, 90 minutes before kickoff	Injury reports (Friday); snap counts show it after the fact
Player speed / tracking MISSING	Direct measure of losing a step	NGS cushion and separation (free, weekly)
Offensive line quality MISSING	Separates a running back's talent from his blocking	Can be built: pressure rate and yards before contact from free data
Official weekly routes MISSING	The cleanest measure of receiver opportunity	Participation proxy (within 1–4 pts for WRs); FantasyData season totals to check it
Salary slate detail 2016–2021 MISSING	DK prices each slate separately; RotoGuru has one price per player per week	Matched FantasyData 99.7% on the overlap week
Coaching / scheme changes MISSING	New offensive coordinator = new usage patterns	Head coach names (schedules). Coordinators are not in any file.

Your own ideas: what else might be missing?

5 · Where an edge could hide

Every idea below is a **hypothesis, not a finding**. The pattern behind all of them: **DraftKings sets salaries days before kickoff, mostly from past points.** An edge exists wherever something important changes *after* the price is set, or sits *under* the points where a price can't see it. The test is always the same: does it predict points *beyond* what the current model already knows, in most seasons, not just one?

Idea	Why salary might miss it	Data you have for it
Vacated opportunity	A starter is ruled out after pricing; his targets and carries go to someone whose salary didn't move	Injuries + depth charts + target share (all on disk)
Role change	A player's routes or snaps jump before his points do. Price follows points, so it lags.	Snaps (on disk); participation routes (free)
Touchdown luck	A player with lots of red-zone and end-zone targets but few touchdowns is due; the reverse is overpriced. TDs swing year to year; opportunity is steadier.	Play-by-play red-zone counts (free); season end-zone targets (FantasyData)
Early-season reputation lag	Week 1–4 prices lean on last year's points, not the efficiency underneath	FantasyData season totals 2021–25; NGS, PFR, FTN (free)
Aging / losing a step	A vet whose points held up but whose yards created, separation, or cushion fell	Age (crosswalk); yards created (FantasyData); cushion (NGS, free)
Rookies and draft capital	No history means the model knows little; salary is a guess	Draft round/pick (crosswalk); combine (free)
Game environment	High over/under games score more at every position (QBs +60% in 52+ totals). Likely priced in already, so it's worth checking how completely.	Vegas lines (in model); points by total (measured 2026-09-15)
Game script	Big favorites run 4 more times a game, *but don't pass less*. Fading their WRs loses (9.8 vs 7.7 pts vs. big underdogs).	Spread (in model); measured 2026-09-15
Coach and referee tendencies	Pace and pass rate differ by coach; penalty rates by crew. Salaries rarely account for either.	Coach and referee columns (schedules, unused); play-by-play (free)
Beat the paid projection	Not an edge by itself; it's the **scoreboard**. If the model can't rank players better than FantasyData, it has no edge yet.	FantasyData projections 2021–2025 (on disk)

Already tested (FEATURES.md §9)

- **Buy low on TEs** after a bad game: best-supported finding in the project. **Don't buy low on RBs** (a dud RB is usually hurt or benched). WRs: neutral.
- **Fade WR1s with a green QB**: your original rule, inverted by the data.
- **Wind** changes play-calling but not RB points. Rain is untested (no data).
- The bounce-back rule may be right in *points per dollar* but invisible in points. That test needs the salary data you now have.

Notes: edge ideas to test

6 · Data quality watch-list

Known problems, so they don't become silent bugs later.

Issue	Detail / what to do
Weekly stats end at 2024	Rebuild 2025 from play-by-play before training on it
Rolling averages reset each season	Week 1 has no player history; last-season features would fill it
Model vs. baseline row counts differ	0.5958 (21,071 rows) vs 0.5791 (16,751 rows). Compare on identical rows before trusting the 0.017 gap.
`green_qb` likely miscounts veterans	Counts starts from the first season *in the data* (`features.py:78`), not a QB's career. A 2016 veteran may look "green" for 6 games.
Snap join is by name + team	Name spelling differences drop rows; the ID crosswalk could fix it
2025 depth charts are daily with no week column	Match snapshot dates to game dates, using the latest snapshot *before* kickoff
FantasyData holiday weeks	Thanksgiving and Saturday games are sometimes missing; 2022 Week 16 salaries differ 7% between exports
FantasyData "accuracy" columns	`TargetAccuracy` and `AccuracyRating` stop being tracked in 2024–25; do not use
FantasyData RB shotgun/under-center YPC	Placeholder values in 2021–22; usable 2023+ only
Film-charted stats disagree between services	Drops and broken tackles differ between FantasyData and PFR; pick one source per stat and stay with it
Projections may have been revised after games	Can't prove FantasyData projections were frozen pre-game; check before claiming the model beats them
No Sunday-morning DK snapshot for 2026 Week 1	Run the slate watcher close to lock each week to capture final statuses